Technical Blueprint:
IMCI Data Pipeline

You have selected the perfect file (WHO IMCI). These are the exact 4 technical steps you need to execute now in your code editor to finish Day 1 of the Hackathon.

The 4 Action Steps

1. Parsing (LlamaParse)

Write a script using `LlamaParse` API to extract the IMCI PDF. Why? Because IMCI is 90% tables (Assess/Classify/Treat). Standard text extraction will destroy this logic. LlamaParse converts these tables into clean Markdown.

2. Chunking (Section-Aware)

Don't split by character count! Write a function to split the Markdown file by Headers (e.g., `# COUGH OR DIFFICULT BREATHING`). Keep chunks around 300-500 tokens. This keeps symptoms and treatments grouped together.

3. Metadata Schema

For every chunk you create, you MUST attach a JSON object containing: `document_name` (IMCI), `section_title` (e.g., Diarrhoea), and `page_number`. This is mandatory for the Citation requirement.

4. Embedding & Chroma DB

Initialize a local `Chroma` client. Pass your chunks through an embedding model (like `OpenAI text-embedding-3` or `sentence-transformers`) and save them to your Chroma collection.

The Metadata Payload

- This is the most critical code structure for Day 1.
- When the AI generates an answer on Day 3, it will read this metadata to tell the doctor exactly where the info came from.
- If you skip this step, your RAG system will fail the "Citation" rule.

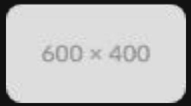
```
// Example Chunk Object in your code
{
  "id" : "imci-chunk-045" ,
  "text" : "If child has stridor in calm state, classify as SEVERE PNEUMONIA..." ,
  "metadata" : {
    "source" : "WHO_IMCI_Handbook.pdf" ,
    "section" : "Assess & Classify Cough" ,
    "page_number" : 14 ,
    "color_code" : "Red (Urgent)"
  }
}
```

Your Next

Immediate Action

- **1. Open VS Code:** Create a new Python project folder.
- **2. Install Packages:** Run `pip install llama-parse chromadb sentence-transformers` in your terminal.
- **3. Load the PDF:** Place `9241546441.pdf` in a `/data` folder inside your project.
- **4. Write the Parser:** Write the first script to send this PDF to LlamaParse and save the output as `imci_parsed.md`.

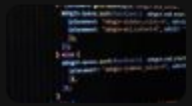
Image Sources



http://googleusercontent.com/image_collection/image_retrieval/3117341918233641364_0

Source: en.idei.club

Image Sources



https://cdn.builtin.com/cdn-cgi/image/f=auto,fit=cover,w=1200,h=635,q=80/sites/www.builtin.com/files/2025-06/dark-mode-coding.jpg

Source: builtin.com